

Detecting Abnormal Markets

Early Warning Systems with Deep & Classical Anomaly Detection

Peng Rao, Maxime d'Aviau de Ternay, Qiao Zhaochen,
Shourya Vashistha, Arnau Nomen

June 1, 2026

Politecnico di Milano

Why this matters

Crashes are rare but decisive – 1929, 2000, 2008, 2020 ...

A handful of days drives a lifetime of returns.

Buy-and-hold S&P 500, 1927–2015:

Strategy	Growth of \$1
Stay invested	\$115.40
Miss the 10 worst days	\$362.01
Miss the 10 best days	\$38.28

Tail events cluster

The frequency and cross-asset correlation of simultaneous heavy losses keep rising.

Diversification fails exactly when you need it most – so we need a signal that fires *before* the drawdown compounds.

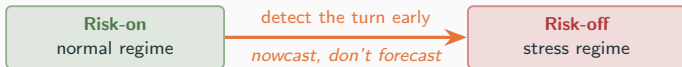
The transition we must catch

Risk-on

Confidence. Equities, credit, EM, commodities bid up. Volatility low, spreads tight, capital chases growth.

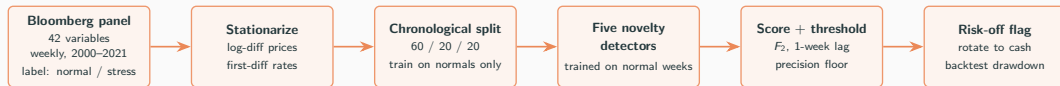
Risk-off

Fear. Flight to cash, short govies, gold, CHF, JPY. Volatility spikes, correlations surge – systemic risk.



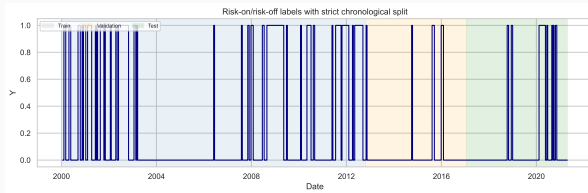
We do not predict *which* crash. We flag, in real time, that this week is “**not normal**” – a 2008-style and a 2020-style stress both qualify.

Pipeline: from raw prices to a risk-off flag

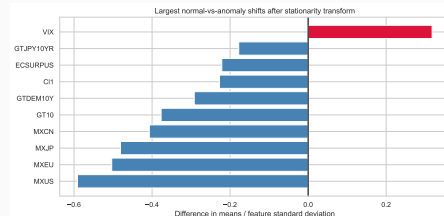


- Treat **normal markets** as the inlier distribution; train on normals only and flag high anomaly scores – no need to label crisis types ex ante.
- Deployment: a flag rotates into cash / short govies / gold. User: CIO, risk officer, wealth manager. **Goal = cut drawdown**, not chase alpha.

Data: stress clusters around drawdowns



US equity (MXUS) with stress weeks shaded



Largest standardized mean shifts, normal vs. stress

1110 weekly observations $\times$ 42 features; **21%** of weeks labelled abnormal. Stress weeks line up with equity selloffs and load on equity returns (MXUS, MXEU, MXJP) and the VIX – the signal is real and economically sensible.

Five detectors, two families

Train on **normal weeks only** $\Rightarrow$ anomaly score $\Rightarrow$ threshold $\Rightarrow$ risk-off flag

Classical / density

Isolation Forest (+ Top- K) – anomalies isolate in fewer random splits

COPOD – empirical-copula tail probabilities, asset-grouped sub-models

Deep generative

GAN – discriminator realism

1 – $D(x)$ & latent reconstruction

VAE – KL-regularised reconstruction error

LSTM-AE – 12-week window, recon + Mahalanobis

One shared objective (F_2) tunes every threshold $\Rightarrow$ comparable operating points across all five.

Honest evaluation: no look-ahead

Train · 60%	CV · 20%	Test · 20%
2000–2012 472 normal weeks	2012–2017 tune threshold	2017–2021 29 crises (13%)

Strict chronological split

- Fit on the *past* only – faithful to live deployment.
- Threshold: all five maximise F_2 ($\beta = 2.0$, recall weighted $4\times$), subject to a precision floor.
- Backtest acts on the signal with a **1-week lag**
 - no same-week look-ahead.

No shuffling, no leakage – numbers are deliberately conservative and deployable.

Metrics for the risk officer

- **Recall / FNR** – a missed crisis is the costly failure.
- **PR-AUC** – honest under heavy class imbalance.
- Precision, F1, ROC-AUC reported for completeness.

Cross-model scoreboard (out-of-sample test)

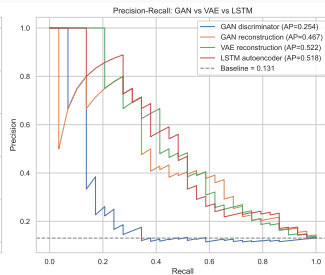
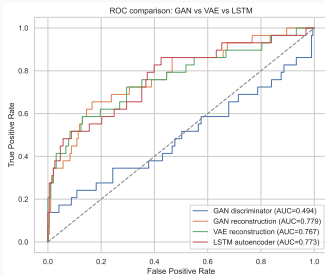
Detector	Caught	Recall	Prec.	F2
Top- <i>K</i> Isolation Forest	24/29	82.8%	58.5%	0.764
Optimized Isol. Forest	14/29	48.3%	58.3%	0.500
COPOD (asset-grouped)	12/29	41.4%	37.5%	0.405
VAE reconstruction	8/29	27.6%	80.0%	0.317
LSTM autoencoder	8/29	27.6%	80.0%	0.317
GAN reconstruction	8/29	27.6%	80.0%	0.317
GAN discriminator	4/29	13.8%	50.0%	0.161

29 true crises in the held-out 2017–2021 test window.

The read

- **No free lunch:** recall vs. precision is a choice, not a ranking.
- **Top-*K* IF** = recall champion: catches 24 of 29 crises.
- **Deep models** = precision champions: fire rarely, right 80% of the time.
- **COPOD** sits in between.

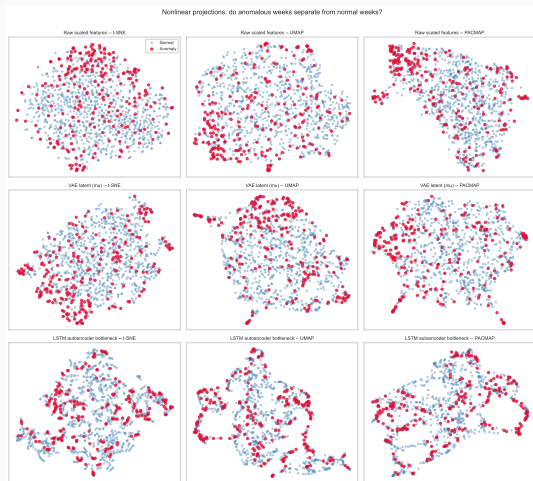
Deep detectors: precise, but cautious



Under the no-leakage split the three reconstruction detectors **converge**: each catches 8/29 crises at 80% precision.

- Best PR-AUC: **VAE 0.52**, LSTM 0.52, GAN-recon 0.47.
- GAN *discriminator* alone is weak (ROC-AUC $\approx$ 0.49).
- Reconstruction beats realism scoring for novelty.

Does the structure actually separate?

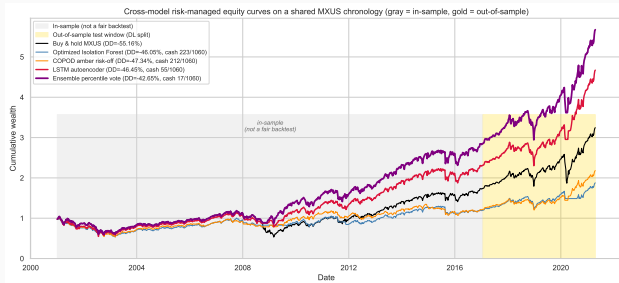


t-SNE / UMAP / PACMAP
on three representations:

- **Raw features** – partial separation only.
- **VAE latent** – barely better.
- **LSTM bottleneck** – clearest cluster; temporal context helps.

Visual proof that adding short-term context sharpens the manifold.

Portfolio impact: every flag cuts drawdown



Test-window backtest

cash when flagged, else MXUS

Strategy	Wealth	Max DD
Buy & hold MXUS	1.87	-27.7%
LSTM autoencoder	2.03	-20.2%
Optimized IF	1.57	-19.7%
COPOD amber	1.80	-18.4%

Signal acts with a 1-week lag (no same-week look-ahead).

- All three **cut drawdown** vs buy & hold ($-27.7\% \rightarrow -18$ to -20%).
- **Only the LSTM** also **grows wealth** above buy & hold; IF/COPOD trade return for deeper drawdown control.

Takeaways

Crises caught

24/29

Top-K Isolation Forest

Best precision

80%

VAE / LSTM / GAN-recon

Beats B&H wealth

2.03

LSTM autoencoder

Min. drawdown

−18.4%

COPOD, vs. −27.7% B&H

- **A strict chronological split** kills look-ahead bias – the numbers are conservative but deployable.
- **No single winner:** pick the detector by mandate. Capital preservation → high-recall Top-K IF; avoid false alarms → high-precision deep models; balanced → COPOD.
- **The robust result:** *every* warning system reduces out-of-sample drawdown (−27.7% → −18 to −20%); the LSTM is the one that also beats buy-and-hold on final wealth.

Bottom line

Framing market stress as novelty detection turns five very different models into one deployable risk-off signal – and the portfolio drawdown reduction survives an honest, leakage-free backtest.